

Quantum-safe three-party lattice based authenticated key agreement protocol for mobile devices

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ABSTRACT

Mobile-based web apps have gained popularity among phone devices because of the rapid advancement of mobile communication and equipment capabilities. Users may utilize their handsets to obtain different types of online services through the Internet at any time and from anywhere. The last three decades have seen widespread use of the three-party AKA protocol, built on asymmetric-key cryptography to produce a private communication channel via any cellular connection between two mobile users. The server can help two mobile users with the assistance of a three-party authentication protocol so that they may create a high-entropy shared private session key. According to several recent studies, all the conventional public-key cryptosystem-based three-party AKA schemes could not withstand attacks due to quantum computing. To get resistance to quantum attacks, Islam and Basu (2021) proposed a new password-based three-party authentication protocol for mobile users using ideal lattices. We improve the efficiency of lattice-based authenticated key exchange protocol and the security of this protocol by making it resistant to password-guessing attacks. A comparative study along with performance and security evaluation is also presented, which demonstrates the applicability of the proposed design.

1. Introduction

The authenticated key agreement protocol, abbreviated as AKA protocol, has been updated by many cryptography experts to provide secure smartphone user communication over any open wireless mobile network. Nevertheless, the huge interconnections of mobiles are easy to utilize. But today, ensuring the privacy and security of messages transferred between mobile users is extremely challenging. The traditional approach to solve these problems is to use the AKA scheme, which enables mobile users to decide a highly unpredictable session key for further secure communication among them [1]. A semantically secure network is created between them by sharing a session key, which enables data security in future communication even if an enemy is eavesdropping on this communication [2].

As quantum computers become a reality, researchers will see two eras in the area of cryptography, the first one is pre-quantum, and the second one is post-quantum. Almost all existing public key encryption techniques of the pre-quantum era will become obsolete in the post-quantum era because of the Shor factoring algorithm proposed in [3,4]. The shor factoring algorithm provides prime factors of a given number in polynomial time on a quantum computer. Because of this algorithm,

encryption techniques like RSA, ElGamal, ECC, etc., will not be secure against quantum attacks [5–7]. These schemes were designed based on the idea given in [8].

For security against quantum computers, people have designed encryption and decryption schemes based on lattices, multivariate polynomials, code, and hashing. These schemes are secure because of some computational problems for which a quantum computer has no known polynomial time algorithm. In effect, the authentication protocols designed in the pre-quantum era should be redesigned so that they must work against quantum computers. This paper focuses on lattice-based schemes [9].

Lattice-based cryptography is widely used primitive to achieve post-quantum security. Lattices have become a highly appealing basis for cryptography during the recent decade. Lattice-based primitives are appealing because their security may frequently be predicated on worst-case hardness assumptions, and they seem to be resistant to quantum computers. In 2013 Lyubashevsky et al. [10] introduced an Ideal lattice and ring learning with errors (RLWE) problem.

Zhang et al. [11] presented an applicable and demonstrably safe two-pass authenticated key exchange mechanism using ideal lattices. In

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the random oracle model, Zhang et al. demonstrated the security of the scheme with low perfect forward secrecy. In transport layer security, Alkim et al. proposed an RLWE-based key exchange protocol [12]. Alkim et al. review the Bos et al. [13] key exchange technique in [12], which is a ring-learning-with-errors (Ring-LWE) based key exchange protocol. Alkim et al. specifically proposed new parameters and better-suited error distribution and introduced a novel and more successful mechanism that significantly increases post-quantum security while requiring a smaller key size. Next, a provably secure password-based AKE protocol based on the RLWE problem was given by Ding et al. [14]. After that, Feng et al. [15] build two lattice-based PAKE (Password Authenticated Key Exchange) protocols using the recently developed approach described in [14], which have a very simple and elegant design. These protocols are a parallel extension of the class of Random Oracle Model (ROM)-based protocols but are used in a lattice-based environment. The signal function from the RLWE key exchange [14] is then used by Ding et al. in 2018 to create an effective zero-knowledge authentication mechanism to authenticate an RLWE key with a secret value and an error element in the Random Oracle Model (ROM) [16].

In recent years, some literature came in three-party lattice-based authentication, but either these articles did not contain security proof or did not publish anywhere. A crucial cryptographic basic is the Three-party Password Authenticated Key Exchange (3PAKE) protocol, which enables entities to create session keys using simple to-remember passwords. In the 3PAKE protocol, it was assumed that the server might not be trustworthy. In 2017, Xu et al. [17] developed the first lattice-based 3PAKE method improving the RLWE-based protocol by Ding et al. [14], although this method is extremely sophisticated because it utilizes the 3PAKE protocol with direct server authentication and the security model they utilized in this study is not intended for a quantum attacker. Using the [17] approach, Rakyong Choi et al. [18] develop a novel lattice-based three-party PAKE protocol with implicit server authentication.

By using smooth projection of the hash function on lattices, Yu et al. [19] designed a new 3PAKE scheme. After that, Liu et al. [20] also gave a provably secure three-party password-based authenticated key exchange protocol based on ring learning with error problems. Shu et al. [21] came up with a 3PAKE scheme constructed on an ideal lattice by adopting the Peikert [22] error reconciliation mechanism. This protocol reduces the space complexity and negotiates a session in six rounds of communications. Yao et al. [23] proposed the first biometric-based privacy-preserving RLWE-based remote biometric authentication scheme for single and multi-server environments, which was an improvement of the work done by Lwamo et al. [24]. For IoT devices, Rana et al. [25] proposed a lattice-based key agreement technique under the RLWE problem. To distinguish between noisy ring multiplications and uniform distributions is the primary goal of the RLWE challenge. Simultaneously, Tang et al. [26] constructed a verifier-based 3PAKE scheme on ideal lattices by using a smooth projection of the hash function technique and a message authentication technique. In the postquantum world mentioned in [15], Feng et al. presented an anonymous authenticated key exchange system for mobile devices. For mobile devices, the protocol is straightforward, elegant, and effective, but it is also susceptible to signal leakage, spoofing, and user anonymity breach attacks. Dabra et al. [27] have suggested a lattice-based anonymous password-authenticated key exchange mechanism for mobile devices to address the aforementioned issues. Especially, [27] said to enable key reuse. Then, Ding et al. demonstrated that when the master key is again utilized in [27], the scheme is susceptible to signal leakage attacks. A stronger protocol was suggested by Ding et al. [1], who also provided formal security proof. The security study demonstrates that the upgraded [1] technique can offer all the security characteristics stated in [27] while being able to withstand a signal leakage attack. Dharminder and Dharminder [28] proposed an RLWE-based anonymous authentication protocol for mobile devices. A provably secure and anti-quantum authentication technique based

on randomized RLWE was proposed by Guo et al. [29]. Based on pre-negotiation and fewer authentication transmissions, the protocol complies with the pre-established security standards and shortens the entire authentication phase latency. For satellite communication, Kumar and Garg [30] presented an authentication system based on the RLWE problem, resistant to quantum assaults.

Recently, Islam and Basu [2] proposed an authenticated key exchange scheme for mobile devices in which a session key was established between two parties using a server. In this scheme, the users who want to communicate, log in first to a server using their password, where the session key gets established between users with the help of a trusted party. Islam and Basu said that their scheme is secure in a post-quantum environment. But, their scheme requires the storage of a masked password and its verifying condition, which may lead to a password guessing attack if the password belongs to a dictionary. In addition, the communicating parties require to perform multiple communications in order to establish a secure session. Hence, there is a need for a secure and more efficient three-party authentication and key agreement scheme.

1.1. Motivations

Numerous authenticated (two and three-party) key agreement protocols based on a certificate, asymmetric-key cryptosystem, and identity-based cryptosystem are available in the literature. These schemes require extra transaction costs and heavy storage for handling users' private and public keys. However, such overheads are not present in the three-party AKA protocols. Only a few AKA protocols proved secure in the random oracle model (ROM). The AKA scheme based on the discrete logarithm problem and the integer factorization problem is found vulnerable to quantum attacks, according to current investigations [3,4]. Ideal lattice is a mathematical concept that has recently gained popularity in constructing safe and robust cryptographic algorithms. The following features make lattice-based cryptographic protocols superior to traditional cryptographic protocols:

- Lattice-based cryptosystems offer attack resistance based on worst-case difficult considerations, whereas conventional cryptosystems offer security based on average-case hardness considerations.
- Schemes based on lattices are robust in the quantum era, whereas the traditional cryptographic techniques are not safe against quantum threats.

1.2. Contribution

Our main contributions to this paper are as follows:

1. Communication and computation costs are better than the previously known protocols based on lattice-based cryptography.
2. The proposed protocol achieves resistance against password-guessing attacks because of additional biometric authentication.
3. The security of the suggested scheme relies on the complex lattice problems, hence it is secure against quantum attacks.
4. A key gets established with only 4 messages exchanged in the authentication phase whereas Islam et al. needs more number of messages transfer.

1.3. Organization

Section 2 contains preliminaries that are required to understand the paper. In Section 3, the proposed scheme is explained. Then in Section 4, formal and informal security analyses of the proposed protocol are done. In the last section, the calculation of the efficiency parameters is done and then the efficiency and security attributes of our scheme is compared with the other schemes.

2. Preliminaries

In this section, we define the required terminologies for the ring learning with error problems which we will use in this work.

2.1. Ring learning with error

We describe some basic knowledge about *RLWE*. We also go through some mathematical problems that can be utilized in the security of the suggested scheme. Let q be a large prime number and $\mathbb{R}$, $\mathbb{Z}$ be the symbol for set of real numbers and integers, respectively. Let $\mathbb{Z}[x]$ and $\mathbb{Z}_q[x]$ be the ring of polynomials over $\mathbb{Z}$ and $\mathbb{Z}_q$ respectively. Let n be a security parameter such that $n = 2^l$ where $l \in \mathbb{Z}^+$. We can define an irreducible polynomial over $\mathbb{Z}$ as $x^n + 1 \in \mathbb{Z}[x]$. Consider a polynomial ring $Q = \mathbb{Z}[x]/\langle x^n + 1 \rangle$ and $Q_q = \mathbb{Z}_q[x]/\langle x^n + 1 \rangle$. Then the elements of ring Q is $a = \sum_{i=0}^{n-1} a_i x^i \in Q$. The L_2 and L_∞ norm are defined as $\|a\|_2 = \{\sum_{i=0}^{n-1} a_i^2\}^{1/2}$ and $\|a\|_\infty = \max_{0 \leq i \leq n-1} |a_i|$ respectively. Let χ_β be the discrete Gaussian distribution over Q_q , where β be any fixed real number which is known as the standard deviation of χ_β . We define some important lemmas as follows.

Lemma 2.1 ([14]). For any $x, y \in Q$ the following inequalities $\|x \cdot y\|_a \leq n^b \|x\|_a \cdot \|y\|_a$ where $(a, b) \in \{(2, 1/2), (\infty, 1)\}$ holds.

Lemma 2.2 ([31,32]). For $\beta = \omega(\sqrt{\log_2 n})$ and $\alpha = \beta\sqrt{n}$, the inequality $\Pr_{d \leftarrow \chi_\beta}[\|d\|_2 > \alpha] \leq \frac{2}{2^n}$ holds.

Let $T = \{-\lfloor \frac{q}{4} \rfloor, \dots, \lfloor \frac{q}{4} \rfloor\}$ be a subset of $\mathbb{Z}_q = \{-\frac{q-1}{2}, \dots, \frac{q-1}{2}\}$. Let z be a variable which varies in the set $\mathbb{Z}_q$. The characteristic function g which takes the input z be defined as:

$$g(z) = \begin{cases} 0 & \text{if } z \in T \\ 1 & \text{if } z \notin T \end{cases}$$

The function $f_2 : \mathbb{Z}_q \times \{0, 1\} \rightarrow \{0, 1\}$ defined by

$$f_2(u, d) = (u + d \cdot \frac{q-1}{2}) \bmod q \bmod 2,$$

where $u \in \mathbb{Z}_q$ and $d = g(u)$ is known as auxiliary modular function.

Lemma 2.3 ([11]). For a given q and the elements $u, e \in Q_q$ such that $|e| < \frac{q}{8}$, we have $f_2(u, g(u)) = f_2(w, g(u))$ where $w = u + 2 \cdot e$.

By the following method, we can easily generalize the functions g and f_2 on Q_q . Given an element $d = d_0 + d_1x + \dots + d_{n-1}x^{n-1} \in Q$, we can also present the vector a as $d = (d_0, d_1, \dots, d_{n-1})$. For vector u define $u = (u_0, u_1, \dots, u_{n-1}) \in \{0, 1\}^n$. The characteristic and auxiliary modular function can be generalized as $g(d) = (g(d_0), g(d_1), \dots, g(d_{n-1}))$ and $f_2(d, u) = (f_2(d_0, u_0), f_2(d_1, u_1), \dots, f_2(d_{n-1}, u_{n-1}))$.

The following *RLWE* problem states that for the given polynomial number of known polynomial pairs $(a, b = a \cdot s + e) \in Q_q \times Q_q$, it is hard to find unknown polynomials $s, e \in Q_q$ chosen from Gaussian distribution.

Definition 2.1 (Ring Learning with Errors (RLWE) Problem). Suppose $A_{s, \chi_\beta} = (u_i, v_i)$ be a sample from $Q_q \times Q_q$ where u_i is taken uniformly from Q_q and $v_i = u_i \cdot s + e_i$, where $s, e \leftarrow \chi_\beta$. RLWE problem is distinguishing elements of A_{s, χ_β} from the uniform distribution on $Q_q \times Q_q$, with only polynomially many samples [10].

The following problems can be reduced to RLWE problems, meaning that if PWE or DPWE problem is solvable in polynomial time, then the RLWE problem can be solved in polynomial time by any quantum computer.

Definition 2.2 (Pairing with Error (PWE) Problems). A function $\gamma : Q_q \times Q_q \rightarrow \{0, 1\}$ is defined by $\gamma(u, s) = f_2(u \cdot s, g(u \cdot s))$. The aim of PWE is for given $u, v, c \leftarrow Q_q$ to find $\gamma(u, s)$ for unknown $s, e' \in \chi_\beta$, where $v = c \cdot s + 2 \cdot e'$.

Table 1

Symbol table.

H	Secure hash function
ID_x	Identity of $x \in \{A, B, S, U\}$
MD_u	Mobile device of U
χ_β	Gaussian distribution
a	$a \in R_q$ publicly known
s	Server S secret key
$H_b(\cdot)$	Biohashing function

Definition 2.3 (Decision Pairing with Error (DPWE) Problem). For a given $d, y, z, c \in Q_q$, the purpose of DPWE problem is to determine whether (d, z) where $d = y \cdot s + 2 \cdot g$ and $z = c \cdot s + 2 \cdot e$ is uniformly random in $Q_q \times Q_q$. Note that $s, g, e \in \chi_\beta$ are unknowns.

3. Proposed scheme

In this part, we go over the four steps of the lattice-based authentication procedures for mobile devices (see Tables 1 and 2).

3.1. Setup phase

Symbols we have used in this study are presented in Table 1. In this phase, S executes the following steps.

1. Selects a random number $a \in Q_q$ and a discrete Gaussian distribution χ_β .
2. Chooses an odd prime number $q > 16\beta^2 n^{3/2}$ and an integer $n \in \mathbb{Z}_q$ that is non-negative power of 2 and satisfying $q \bmod 2n = 1$.
3. Selects a hash function which is defined as $H : \{0, 1\}^* \rightarrow \{0, 1\}^l$ where l is a fixed length output.
4. Publishes $\{n, q, a, \chi_\beta, H(\cdot), H_b(\cdot)\}$ in a public domain and preserves the secret key s confidentially.

3.2. Registration phase

1. The user U selects his/her identity ID_u , password PW_u and generates a random number a_u , and then computes $MPW_u = H(ID_u, PW_u, a_u)$. After that U sends the message $\langle ID_u, MPW_u \rangle$ to S .
2. After receiving the message from U , the server S computes $S_u = H(ID_u, s)$, $D_u = S_u \oplus MPW_u$ and $V_u = H(ID_u, MPW_u, D_u)$. Finally, S sends the message $\langle D_u, V_u \rangle$ to U . S also stored the registered identities corresponding to T_R , where T_R is the time of registration.
3. The user U imprints Bio_u , computes $H_b(Bio_u)$ and $MA_u = H_b(Bio_u) \oplus a_u$, and finally Stores (MA_u, D_u, V_u) in MD_u . The steps of registration process are described in Table 2.

3.3. Authentication phase

When a user A would like to establish a secure session with another user B , he/she would be able to complete this task with the help of a trusted server. Our order of communications in this phase is shown in Table 3. The process of secure session establishment is as follows. Note that if any of the mentioned verification becomes unsuccessful the session gets terminated.

1. A inputs (ID_A, PW_A) into his/her MD_A and imprints Bio_A . MD_A computes $a_A = MA_A \oplus H_b(Bio_A)$, $MPW_A = H(ID_A, PW_A, a_A)$, and $S_A = D_A \oplus MPW_A$. Finally, MD_A verifies $V_A = H(ID_A, MPW_A, S_A)$. If the verification is successful, then A samples $r_A, f_A \xleftarrow{R} \chi_\beta$ and computes $x_A = a \cdot r_A + 2 \cdot f_A$. After that, MD_A performs $\Sigma_1 = H(ID_A, ID_B, S_A, x_A, T_A)$. Finally, MD_A sends the message $\langle ID_A, ID_B, x_A, \Sigma_1, T_A \rangle$ to B , where T_A is a current timestamp

Table 2

User registration.

U	Secure channel	S
Selects ID_u, a_u, PW_u Computes $MPW_u = H(ID_u, PW_u, a_u)$	$\langle ID_u, MPW_u \rangle$	Computes $S_u = H(ID_u, s)$ $D_u = S_u \oplus MPW_u$ $V_u = H(ID_u, MPW_u, D_u)$
Input Bio_u and computes $H_b(Bio_u)$ and $MA_u = H_b(Bio_u) \oplus a_u$ Stores (MA_u, D_u, V_u) in mobile device	$\langle D_u, V_u \rangle$	

Table 3

Authentication through public channel.

A	B	S
Input ID_A and PW_A Imprint Bio_A & Computes $a_A = MA_A \oplus H_b(Bio_A)$ $MPW_A = H(ID_A, PW_A, a_A)$ $S_A = D_A \oplus MPW_A$ Verify $V_A \stackrel{?}{=} H(ID_A, MPW_A, S_A)$ If verification of credentials holds, Selects $r_A, f_A \xleftarrow{R} \chi_\beta$ $x_A = a \cdot r_A + 2 \cdot f_A$ $\Sigma_1 = H(ID_A, ID_B, S_A, x_A, T_A)$ $\langle ID_A, ID_B, x_A, T_A, \Sigma_1 \rangle$	Checks the freshness of T_A Input ID_B and PW_B Computes $a_B = MA_B \oplus H_b(Bio_B)$ $MPW_B = H(ID_B, PW_B, a_B)$ $S_B = D_B \oplus MPW_B$ Verify $V_B \stackrel{?}{=} H(ID_B, MPW_B, S_B)$ On success of verification, Selects $r_B, f_B \xleftarrow{R} \chi_\beta$ $x_B = a \cdot r_B + 2 \cdot f_B$ $t_B = x_A \cdot r_B$ $w_B = g(t_B)$ Computes $\Sigma_2 = H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A)$ $\langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$	Checks the freshness of Timestamp T_A Computes $S_A = H(ID_A, s)$ $S_B = H(ID_B, s)$ Verify $\Sigma_1 \stackrel{?}{=} H(ID_A, ID_B, S_A, x_A, T_A)$ $\Sigma_2 \stackrel{?}{=} H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A)$ If both verification holds, computes $\Sigma_3 = H(ID_A, ID_B, S_A, x_A, x_B, w_B, T_A)$ $\Sigma_4 = H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A, \Sigma_3)$ $\langle \Sigma_3, \Sigma_4 \rangle$
Computes $\Sigma_3 = H(ID_A, ID_B, S_A, x_A, x_B, w_B, T_A)$ $t'_B = x_B \cdot r_A$ $k'_B = f_2(t'_B, w_B)$ $Sk_{AB} = H(ID_A, ID_B, x_A, x_B, k'_B, T_A)$ Verify $\Sigma_5 \stackrel{?}{=} H(Sk_{AB}, \Sigma_3, T_A)$	Verify $\Sigma_5 \stackrel{?}{=} H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A, \Sigma_3)$ Computes $k_B = f_2(t_B, w_B)$ $Sk_{BA} = H(ID_A, ID_B, x_A, x_B, k_B, T_A)$ Computes $\Sigma_5 = H(Sk_{BA}, \Sigma_3, T_A)$ $\langle x_B, w_B, \Sigma_5 \rangle$	

- The user B checks the freshness of timestamp and inputs (ID_B, PW_B) into MD_B and imprints Bio_u . Then MD_B computes $a_B = MA_B \oplus H_b(Bio_B)$, $MPW_B = H(ID_B, PW_B, a_B)$, and $S_B = D_B \oplus MPW_B$. Finally, MD_B verifies $V_B = H(ID_B, MPW_B, S_B)$. If the verification is successful, B samples $r_B, f_B \xleftarrow{R} \chi_\beta$ and computes $x_B = a \cdot r_B + 2 \cdot f_B$, $t_B = x_A \cdot r_B$, $w_B = g(t_B)$, and $\Sigma_2 =$

- $H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A)$. Then, MD_B sends the message $\langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$ to S .
- As the server S receives the message $\langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$, S checks whether ID_A and ID_B are registered or not. If both are registered users, S checks the freshness of timestamp T_A . After every successful attempt, S update previous timestamp with fresh timestamp.

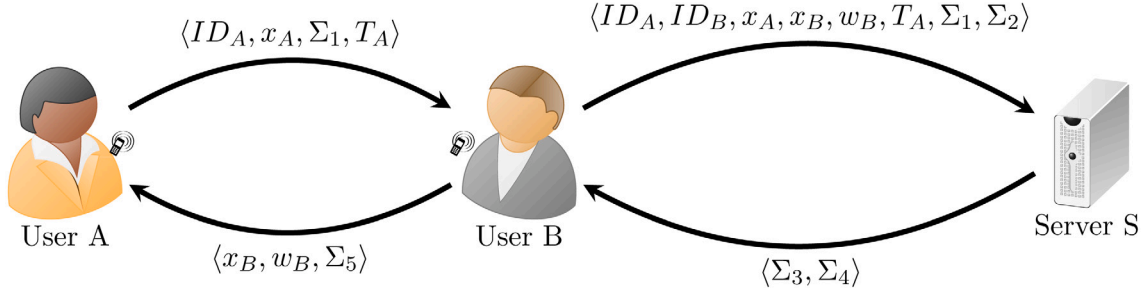


Fig. 1. The content of messages and the order of communications in our scheme.

Table 4

Password update phase.

U	Mobile device of U
$\langle ID_u, MPW_u, Bio_u \rangle$ to the mobile device	$S_u = h(ID_u, s)$ $D'_u = S_u \oplus MPW'_u$ $V'_u = H(ID_u, PW'_u, D'_u)$ if $V'_u \neq V_u$ and $MA_u \neq H_b(Bio_u) \oplus a_u$, then terminate the request. Otherwise, ask the ID_u to enter new password.
Selects PW_{in} in MD_u	
Computes $MPW_{in} = H(ID_u, PW_{in}, a_u)$	
$\langle ID_u, MPW_{in} \rangle$ to mobile device	$S_u = h(ID_u, s)$ $D_{in} = S_u \oplus MPW_{in}$ $V_{in} = H(ID_u, PW_{in}, D_{in})$ Replace $\{MA_u, D_u, V_u\}$ by $\{MA_u, D_{in}, V_{in}\}$

- The server S computes $S_A = H(ID_A, s)$ and $S_B = H(ID_B, s)$. Then the server S verifies whether $\Sigma_1 \stackrel{?}{=} H(ID_A, ID_B, S_A, x_A)$ and $\Sigma_2 \stackrel{?}{=} H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A)$. If the both verifications are successful, then the server S computes $\Sigma_3 = H(ID_A, ID_B, S_A, x_A, x_B, w_B, T_A)$ and $\Sigma_4 = H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A, \Sigma_3)$. After that the server sends the message $\langle \Sigma_3, \Sigma_4 \rangle$ to B .
- The user B verifies whether $\Sigma_4 \stackrel{?}{=} H(ID_A, ID_B, S_B, x_B, x_A, w_B, T_A, \Sigma_3)$. If the verification is successful, then B computes $k_B = f_2(t_B, w_B)$, $Sk_{BA} = H(ID_A, ID_B, x_A, x_B, k_B, T_A)$ and $\Sigma_5 = H(Sk_{BA}, \Sigma_3, T_A)$. Finally, B sends the message $\langle x_B, w_B, \Sigma_5 \rangle$ to A .
- As A receives the message from B , it computes $\Sigma_3 = H(ID_A, ID_B, S_A, x_A, x_B, w_B, T_A)$. After that A computes $t'_B = x_B \cdot r_A$, $k'_B = f_2(t'_B, w_B)$, and $Sk_{AB} = H(ID_A, ID_B, x_A, x_B, k'_B, T_A)$ and then verify whether $\Sigma_5 \stackrel{?}{=} H(Sk_{AB}, \Sigma_3, T_A)$. If the verification holds, then the authentication and key exchange is successful. The content of messages and the order of communications in our scheme presented in Fig. 1.

3.4. Password change phase

- The user U inputs his/her ID_u , PW_u and Bio_u into MD_u . The mobile device of U computes $S_u = h(ID_u, s)$ and $D'_u = S_u \oplus MPW'_u$ and $V'_u = h(ID_u, PW_u, D'_u)$. Then the user gives his own biometric. If $V'_u \neq V_u$ or $MA_u \neq H_b(Bio_u) \oplus a_u$, then the MD_u terminate the password change request, else ask U for a new PW_u .
- Let U inputs his new password, which is denoted by PW_{in} into his/her MD_u . It computes $MPW_{in} = H(ID_u, PW_{in}, a_u)$ and then the user sends the message $\langle ID_u, MPW_{in} \rangle$ to the device. This device of user U computes $S_u = h(ID_u, s)$, $D_{in} = S_u \oplus H(ID_u, PW_{in}, a_u)$ and $V_{in} = H(ID_u, PW_{in}, D_{in})$. Next, the mobile

device of U replace the number (D_u, V_u, MA_u) with the new values (D_{in}, V_{in}, MA_u) . The steps for password update are described in Table 4.

3.5. Proof of correctness

In our protocol, if $k_B \neq k'_B$ then the user A cannot be able to compute session key correctly. We prove that $k_B = k'_B$ holds. We know that

$$t'_B = x_B \cdot r_A = a \cdot r_B \cdot r_A + 2 \cdot f_B \cdot r_A, \quad (1)$$

$$t_B = x_A \cdot r_B = a \cdot r_A \cdot r_B + 2 \cdot f_A \cdot r_B. \quad (2)$$

After subtracting (2) from (1), we get

$$t_B - t'_B = 2 \cdot (f_A \cdot r_B - f_B \cdot r_A). \quad (3)$$

According to Lemma 2.1, we have:

$$\|f_B \cdot r_A - f_A \cdot r_B\|_2 \leq \|f_B \cdot r_A\|_2 + \|f_A \cdot r_B\|_2 < \sqrt{n} \|f_B\|_2 \cdot \|r_A\|_2 + \sqrt{n} \|f_A\|_2 \cdot \|r_B\|_2 \quad (4)$$

$$< \sqrt{n} \cdot \beta^2 \cdot n + \sqrt{n} \cdot \beta^2 \cdot n \quad (5)$$

$$= 2 \cdot \beta^2 \cdot n^{\frac{3}{2}} \quad (6)$$

$$< q/8 \quad \text{because } q > (4\beta n^{\frac{3}{4}})^2 \quad (7)$$

Using Lemma 2.3, it can be concluded that

$$k'_B = f_2(t'_B, w_B) = f_2(t'_B, g(t_B)) = f_2(t_B, g(t_B)) = k_B. \quad (8)$$

4. Security analysis

In this section, we describe adversary model, present the formal security proof and then discuss the informal security analysis.

4.1. Adversary model

In Random Oracle Model of hash functions, we will have an oracle that will give an answer for every hash input. But once the oracle has given the answer y for the input x then the oracle will always answer y for the input x . If oracle does not know the answer of a query let us say x then the oracle randomly select an answer let us say y from the range space of hash function, stores pair (x, y) , and gives y as the answer for the query x .

Let $\mathbb{U}$ be a set of all possible users and $\mathbb{S}$ be a set of servers. Multiple users (honest or malevolent) $U \in \mathbb{U}$ and a single server $S \in \mathbb{S}$ will be part of our scheme. The communication channel among users and server can be controlled by adversary $\mathfrak{R}$. The adversary can perform various experiments (modify, delete, replace, etc.) with the exchanged messages through the channel. The server S chooses a random secret key from $\mathbb{Z}_q$. The users $U \in \mathbb{U}$ maintains a uniformly chosen password PW_U from a finite-sized dictionary $D \leq 10^6$.

We denote the i th instance (in the i th session) of an entity Q be denoted as Q^i . The adversary $\mathfrak{R}$, and Q communicate by accomplishing

the queries of the oracle mentioned below to entitle the power of $\mathfrak{R}$ to violate the well-formed security of the Sk accepted by the entities of our scheme. The adversary $\mathfrak{R}$ may get the stored information on the mobile by stealing the mobile and may get password PW_u either by guessing attack or some other way. To infringe the security of our scheme, the adversary $\mathfrak{R}$ and challenger C plays a two-player game. This game permits C to implement the suggested scheme and to follow a series of queries which executes in polynomial time. Below these queries are explained in detail:

1. $Execute(\Pi_P^a, \Pi_Q^b)$: If Π_P^a and Π_Q^b have not been used then this query sends all the exchanged messages to PPT adversary $\mathfrak{R}$ according to the protocol.
2. $Send(\Pi_P^a, m)$: The message m will be sent to Π_P^a from $\mathfrak{R}$ and then Π_P^a replies the truthful message to $\mathfrak{R}$ according to the protocol.
3. $Corrupt(U, w)$: The reply of this query is PW (password of U), if $w = 0$ and it returns secret value stored in mobile device, if $w = 1$.
4. $Reveal(\Pi_P^i)$: By sending this query the PPT adversary $\mathfrak{R}$ obtains the session key Sk such that Π_P^i and his partner enter into the state of acceptance.
5. $Test(\Pi_P^a)$: As Π_P^a obtain this query, P chooses bit b at random, if $b=0$ then it choose a binary string at random and sends that string as the reply, if $b=1$ then it eliminates the $Reveal$ query and reply with session key Sk . Keep in mind that only the adversary $\mathfrak{R}$ is permitted to send its partner a test query. The semantic security of session key is defined by this query.

The above queries can be issued only polynomially many times to any random oracle by any adversary.

We say Π_Q^i as *accepted* if it receives the last expected message and then enter into the state of acceptance. Some messages will be sent by Π_Q^i and some other messages will be received by Π_Q^i . The *session identifier* (sid) of Π_Q^i defined as the group of all such messages.

Definition 4.1 (Freshness of a Session). A case Π_X^i where $X \in \{A, B, S\}$ is said to be fresh if Π_X^i is in accepted state, no $Reveal(\Pi_X^i)$ or $Send(\Pi_X^i, m)$ queries can be issued, and only $\langle Corrupt(X), Corrupt(S) \rangle$ query can be issued.

Definition 4.2 (Partner). Two entity instances Π_A^i and Π_B^i are called partners iff

1. They possess the common SID (session identifier).
2. Each of them eventually achieve acceptance, which means, they have established a shared session key, Sk and have validated the other party's identity.
3. Π_A^i and Π_B^i are partners and vice-versa.

Definition 4.3 (Advantage and Semantic Security). Let $\mathfrak{R}$ be an PPT adversary and the success of $\mathfrak{R}$ be the event when $\mathfrak{R}$ correctly predict the bit b chosen in the "Test" query. The advantage of $\mathfrak{R}$ with t as an upper bound of time be defined as

$$Adv_{\mathfrak{R}}(t) = \left| pr(S) - \frac{1}{2} \right|$$

where $pr(S)$ denotes the probability of success of the PPT adversary $\mathfrak{R}$. A scheme is semantically secure if Π_i^P and its partner conclude with accept state and produced the common session key, and the advantage of PPT adversary, $Adv_{\mathfrak{R}}(t)$ is negligible.

4.2. Provable security

By using [33], we proof the security of our proposed scheme.

Theorem 4.1. Let $Adv_{\mathfrak{R}}^{RLWE}(t')$ be the advantage of $\mathfrak{R}$ to resolve an instance of the ring learning with error problem within the time t' , D be the finite sized dictionary of password. The adversary $\mathfrak{R}$ may ask q_H , q_{send} and q_{exec} number of hash, send and execute queries respectively to oracles. Let the term T_{smul} be defined as the cost, i.e., time of component wise multiplication with a scalar in Q_p , and $Adv_{\mathfrak{R}}^{Our\ scheme}(t)$ be the advantage of $\mathfrak{R}$ to compute a Sk in our scheme within the time t . We get

$$Adv_{\mathfrak{R}}^{Our\ scheme}(t) \leq \frac{q_H^2}{2^{l+1}} + \frac{(q_{send} + q_{exec})^2}{2^{l+1}} + (q_{exec} + q_{send}) Adv_{\mathfrak{R}}^{RLWE}(t') + \max\left\{\frac{q_{send}}{|D|}, \frac{1}{q^2} + \frac{q_H^2}{2^{l+1}}\right\}$$

where $t' \leq t + (2q_{exec} + 4q_{send} + 4q_H + 1)T_{smul}$.

Proof. Let C be a challenger which simulates our scheme and $\mathfrak{R}$ be a probabilistic poly time adversary. In the game of $\mathfrak{R}$ and C , $\mathfrak{R}$ can get answer from C by sending the polynomially many series of *Corrupt*, *Send*, *Execute*, *Test* and *Reveal* queries to C . According to our interpretation, C calculates various messages and responds to $\mathfrak{R}$. The adversary $\mathfrak{R}$ will seek to break the well-formed secrecy of Sk calculated between A and B , as per the messages received from C . In this explanation, we are trying to prove that if $\mathfrak{R}$ can fake the Sk , then $\mathfrak{R}$ will also resolve a case of RLWE problem in $Q_q \times Q_q$.

Here, we take the succeeding games $\mathbb{G}_0, \mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_3$ and $\mathbb{G}_4$. The implementation of $\mathbb{G}_0$ is identical to carrying out the violation of the well-formed secrecy of Sk of our scheme using the brute-force method. In every $\mathbb{G}_i$, we come up with Y_i an event ($0 \leq i \leq 4$) using which $\mathfrak{R}$ seeks to infringe the well-formed secrecy of the Sk calculated during the implementation of $\mathbb{G}_i$. Let us take the event E which is free from Y_i , may happen during the implementation of our scheme by $\mathfrak{R}$ including E is distinguishable by C . Focus that $\mathbb{G}_i$ and $\mathbb{G}_{i+1}$ seems similar unless E happens. We get the following inequality that is given in [33].

$$|P[Y_{i+1}] - P[Y_i]| \leq P[E]$$

$\mathbb{G}_0$: The implementation of $\mathbb{G}_0$ is attack of the scheme in the ROM. So, we have:

$$Adv_{\mathfrak{R}}^{Our\ scheme}(t) = |P[Y_0] - 1/2| \quad (9)$$

$\mathbb{G}_1$: The implementation of $\mathbb{G}_1$ is similar to $\mathbb{G}_0$ other than $\mathfrak{R}$ perform Hash queries in a changed manner respective to $\mathbb{G}_0$. Here, C sustains initial empty list H^{list} . This list contains tuples of the format (x, y) , where x is the input to the oracle and y is the respective output. If $\mathfrak{R}$ enquire a query for input x , C examines the list H^{list} and returns y to $\mathfrak{R}$ if a record (x, y) was established in H^{list} . The real attack in the random oracle model and the implementation of this game are identical. Consequently, we get:

$$P[Y_1] = P[Y_0] \quad (10)$$

$\mathbb{G}_2$: The implementation of $\mathbb{G}_2$ is similar as $\mathbb{G}_1$ except that it will be terminated if there is a crash occur in the implementation of $\langle ID_A, x_A, T_A, \Sigma_1 \rangle, \langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$ and $\langle x_B, w_B, \Sigma_5 \rangle$. The event crash occurs if the oracle H returns the identical output for two un-identical inputs. According to the notion of the birthday attack, chances of getting response of H by crashing can be maximum $\frac{q_H^2}{2^{l+1}} + \frac{(q_{send} + q_{exec})^2}{2^{l+1}}$. Consequently, we get

$$|P[Y_2] - P[Y_1]| \leq \frac{q_H^2}{2^{l+1}} + \frac{(q_{send} + q_{exec})^2}{2^{l+1}} \quad (11)$$

$\mathbb{G}_3$: Here, the implementation is similar as $\mathbb{G}_2$ except that the adversary $\mathfrak{R}$ performs H for the load $(ID_A, ID_B, x_A, x_B, k_B, T_A)$. In the implementation of H , the probability that $\mathfrak{R}$ will pick the right bit of $Test$ query is upper bounded by $\frac{q_H^2}{2^{l+1}}$. Consequently, we get

$$|P[Y_3] - P[Y_2]| \leq \frac{q_H^2}{2^{l+1}} \quad (12)$$

$\mathbb{G}_4$: Here, we consider numerous off-line and online password guessing attacks, i.e., the Sk may be calculated without using the hash function; the possible passwords of A and B may be computed in an off-line or online way.

- Sk may be predicted without implementing H , i.e., Sk is completely free from the response of H and $(ID_A, ID_B, x_A, x_B, x_B, k'_B, T_A)$. As per our protocol, Sk is equal to $H(ID_A, ID_B, x_A, x_B, x_B, k'_B, T_A)$ where $k_B = f_2(t_B, w_B)$. Accordingly, we can state that $\mathfrak{R}$ can opportunely predict Sk , and consequently $\mathfrak{R}$ can solve a case of RLWE problem inside the polynomial amount of time t' . For that reason, we get

$$|P[Y_4] - P[Y_3]| \leq (q_{exec} + q_{send}) Adv_{\mathfrak{R}}^{RLWE}(t'). \quad (13)$$

- If $\mathfrak{R}$ does not implement H with the accurate load $(ID_A, ID_B, x_A, x_B, x_B, k'_B, T_A)$, then $\mathfrak{R}$ cannot differentiate the real Sk and a random Sk i.e., $Pr[Y_4] = 1/2$.
- The probability of predicting the password of A and B from D via any online mean is $\frac{q_{send}}{|D|}$.
- To predict password of X where $X \in \{A, B\}$ in off-line mode, the adversary $\mathfrak{R}$ enquires the query $\langle \text{Corrupt}(X), \text{Corrupt}(S) \rangle$ to obtain the data $\langle MA_X, MPW_X \rangle$ from the mobile device of the user X . The probability to predict the password of X where $X \in \{A, B\}$ including the private key of the server S is $\frac{1}{q^2} + \frac{q_H^2}{2^{l+1}}$.

Therefore, we have

$$Pr[Y_4] = \frac{1}{2} + \max\left\{\frac{q_{send}}{|D|}, \frac{1}{q^2} + \frac{q_H^2}{2^{l+1}}\right\}. \quad (14)$$

Adding the Eqs. (9)–(14), we have

$$Adv_{\mathfrak{R}}^{Our\ scheme}(t) \leq \frac{q_H^2}{2^{l+1}} + \frac{(q_{send} + q_{exec})^2}{2^{l+1}} + (q_{exec} + q_{send}) Adv_{\mathfrak{R}}^{RLWE}(t') + \max\left\{\frac{q_{send}}{|D|}, \frac{1}{q^2} + \frac{q_H^2}{2^{l+1}}\right\}$$

$$\text{where } t' \leq t + (2q_{exec} + 4q_{send} + 4q_H + 1)T_{smul}. \quad \square$$

4.3. Informal security analysis

In this part, we examined how well the proposed protocol can withstand various cryptographic attacks.

4.3.1. Resistance to user impersonation attack

As mentioned in the phase of authentication, the server authenticates users by calculating $\Sigma'_1 = H(ID_A, ID_B, S_A, x_A, T_A)$, $\Sigma'_2 = H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A)$ and comparing it with Σ_1 and Σ_2 obtained from A and B . To pretend to be A and B , $\mathfrak{R}$ computes Σ_1 and Σ_2 for this purpose, $\mathfrak{R}$ needs S_A, S_B , and S_A, S_B itself contains the server secret key, which is likewise inaccessible to $\mathfrak{R}$. Hence, $\mathfrak{R}$ cannot compute a legal Σ_1 and Σ_2 . Accordingly, our purposed scheme is resistant to user impersonation attacks.

4.3.2. Resistance to server impersonation attack

In the authentication phase, user A and B authenticates the server by calculating $\Sigma'_3 = H(SK_{AB}, \Sigma_3, T_A)$, and $\Sigma'_4 = H(ID_A, ID_B, S_B, x_A, x_B, w_B, T_A, \Sigma_3)$, and comparing it with Σ_5 and Σ_4 obtained from server. If an adversary wants to pass the comparison by pretending to be the server, they must construct Σ_5 and Σ_4 . However, Σ_5 and Σ_4 contain the server secret key, which is hidden from $\mathfrak{R}$. Hence, $\mathfrak{R}$ cannot compute a legal Σ_5 and Σ_4 . Consequently, our purposed scheme is resistance to server impersonation Attacks.

4.3.3. Resistance to known session key attack

With the support of k_A, t_A, k_B , and t_B the session key $Sk_{AB} = H(ID_A, ID_B, x_A, x_B, k_B, T_A)$ is computed. t_A, t_B is computed using the random numbers r_A, r_B, f_A, f_B . Therefore, even if an adversary has an old session key, he or she will be unable to access the present session key. To put it another way, the suggested approach is protected against the well-known session key attack.

4.3.4. Resistance to replay attack

To authenticate the freshness of sent messages, our proposed protocol stores the current timestamp T_A at the registration step. As a result, the receiver may identify an old message even if it is replayed by checking the freshness of the timestamp. In other words, the method defends replay attacks and supports the freshness of communications.

4.3.5. Perfect forward secrecy

Our scheme achieves total forward secrecy if it prohibits an adversary from gaining the session key by accessing the long-term key, such as secret keys or the names of the entities involved in the system. Our system creates a session key Sk_{AB} that cannot be simply calculated or guessed by an attacker. Sk_{AB} consisting x_A, x_B generated with the help of r_A and r_B chosen uniformly at random from χ_β in the relevant session. As a result, our approach fully supports forward secrecy.

4.3.6. Resistance to key control property

The “no key control” characteristic of an authenticated key exchange protocol indicates that the session key must be decided upon jointly by the two users. The Sk_{AB} is jointly calculated by A and B in the proposed protocol as $Sk_{AB} = H(ID_A, ID_B, x_A, x_B, k_B, T_A) = H(ID_A, ID_B, x_A, x_B, k'_B, T_A)$, where $x_A = a \cdot r_A + 2 \cdot f_A$, and $x_B = a \cdot r_B + 2 \cdot f_B$. Therefore, A cannot be forced by B , to use a preselected value or a value from a limited set. Therefore, no key control attribute is provided by the proposed protocol.

4.3.7. Session key integrity

In the session key integrity property, the adversary $\mathfrak{R}$ must not be able to do anything in the calculation of a session key. Additionally, a session key $Sk_{AB} = H(ID_A, ID_B, x_A, x_B, k_B, T_A)$ computed between the user's A and B must depend exclusively on the input $x_B = a \cdot r_B + 2 \cdot f_B$, generated with the help of r_A, f_A, f_B , and r_B chosen uniformly at random from χ_β in the relevant session. As a result, our approach fully supports session key integrity.

4.3.8. Resistance to privileged-insider attack

After a conversation between A and B has been completed, A believes that the Sk_{AB} has been calculated with B ; nevertheless, B believes that the session key other than A has been created with $\mathfrak{R}$.

4.3.9. Resistance to key replicating attack

In a key replicating attack, an active adversary $\mathfrak{R}$ alerts the captured messages sent and received by A, B , and S during a conversation to force the user's A and B to compute a session key that is not legitimate. But, for modification in the authentication phase of the purposed scheme first of all $\mathfrak{R}$ creates the message $\langle ID_A, ID_B, x_A, T_A, \Sigma_1 \rangle$, $\langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$, $\langle \Sigma_3, \Sigma_4 \rangle$, $\langle x_B, w_B, \Sigma_5 \rangle$ where $\Sigma_1, \Sigma_2, \Sigma_5, \Sigma_4$ includes the secret value of server, which is also hidden from Re . Hence, $\mathfrak{R}$ cannot compute a legal Σ_5 and Σ_4 . Consequently, our purposed scheme is Resistance to Key replicating attacks.

4.3.10. Resistance to men in middle attack

The adversary cannot be able to successfully modify the messages $\langle ID_A, ID_B, x_A, T_A, \Sigma_1 \rangle$ and $\langle x_B, w_B, \Sigma_5 \rangle$ because Σ_1 and Σ_5 use S_A which is known to the server S . Similarly, the adversary cannot be able to successfully modify the messages $\langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$ and $\langle \Sigma_3, \Sigma_4 \rangle$ because S_A and S_B are only known to the server S .

5. Performance analysis

This section compares the performance of execution and communication costs of our suggested three-party authentication protocol with existing schemes. According to the calculations made in [14,15], to uniformly select samples randomly from χ_β , we pick $\log_2 \beta = 17.1$. We use the SHA-512 bits hash function for hashing, which produces a 512 bits digest value. Additionally, we assumed that n, p , and a component of Q_p have sizes of 1024 bits, 12289 bits, and 4096 bits, respectively. The timestamp and identity of the user are of size 32 bits each.

Table 5
Definitions and execution time in nano seconds taken from [2].

Symbol	Definition	Computation cost	
		User	Server
T_β	Cost of sampling from χ_β	561.483	0.73503
T_{sm}	Cost of scalar multiplication (component wise) in Q_q	6.655	0.298
T_{mp}	Cost of multiplication (component wise) in Q_q	13.052	0.307
T_{mpa}	Cost of multiplication (component wise) and addition in Q_q	29.505	2.549
T_{cha}	Cost of computing g function	35.515	0.698
T_h	Cost of computing hash function	180.964	14.09

Table 6
Definitions and execution time of non-lattice-based cryptographic operations in nano seconds taken from [2].

Symbol	Definition	Computation cost	
		User	Server
T_{bp}	Cost of bilinear pairing operation in $\Omega \times \Omega$	52533000	8262000
T_{exp}	Cost of raising power on a number in Ω	3437000	417000
T_{sm}	Cost of multiplying with a scalar in Ω	17909000	2578000
T_{pa}	Cost of adding two points in Ω	78000	12000
T_c	Cost of computing extended chaotic map	32400000	5583000

5.1. Execution cost

In this section, we compare the execution cost of our proposed scheme with recently existing schemes. The Lattice Crypto [34] and MIRACL Libraries [35] are used to build our protocol. The research in [15] has taken the system: a Dell desktop PC (Windows ten Professional Operating System, a three-point four Giga Hertz Intel(R) Core(TM) i7-6700 CPU, and eight Gigabyte Random Access Memory), which considered as a system of server and a Samsung GT-I9300 mobile phone (Android four point three Operating System, a one point four Giga Hertz Quad-core Samsung Exynos 4412 processor, and one Giga Byte Random Access Memory), which considered as specifications of user's device. In Table 5, we declare the running time in nanoseconds of different types of cryptographic operations to calculate our proposed scheme. We ignore the cost of the f_2 function.

In our authentication phase, A selects 2 gaussian random samples in χ_β , executes one-time bio hashing function, six-time hash functions, one-time component-wise multiplication in Q_p , one-time component-wise multiplication and addition in Q_p , one-time scalar multiplication in Q_p . Thus, the total time elapsed is $6T_h + T_{H_b} + 2T_\beta + T_{sm} + T_{mpa} + T_{mp} = 2438.926$ ns. The user B selects two random samples in χ_β , executes a one-time bio hashing function, six-time hash functions, one time T_{ma} operation in Q_p , one-time scalar multiplication in Q_p , one-time component-wise multiplication in Q_p , one-time g function. Therefore, the total cost is $6T_h + T_{H_b} + 2T_\beta + T_{sm} + T_{mpa} + T_{mp} + T_{cha} = 2474.441$ ns. The server S executes the hash functions six times. The total computation cost is $6T_h = 84.54$ μs . Hence, the total execution cost of our protocol is 4913.367 ns.

In Islam and Basu's authentication scheme [2], A selects two gaussian random samples in χ_β , executes eight-time hash functions, one-time component-wise multiplication with addition in Q_p , one-time component-wise scalar multiplication in Q_p , one-time component-wise multiplication in Q_p , one time g function. Hence the total cost is $8T_h + 2T_\beta + T_{sm} + T_{mpa} + T_{mp} + T_{cha} = 2655.405$ ns, B selects two gaussian random samples in χ_β , executes eight-time hash functions, one-time component-wise multiplication with addition in Q_p , one-time component-wise scalar multiplication in Q_p , one-time component-wise multiplication in Q_p , one time g function hence the total cost is $8T_h + 2T_\beta + T_{sm} + T_{mpa} + T_{mp} + T_{cha} = 2655.405$ ns, the S computes six times the hash functions. Thus the execution time is $6T_h = 84.54$ ns. Hence the total execution cost of Islam and Basu scheme [2] is $2655.405 + 2655.405 + 84.54 = 5395.35$ ns.

In Farash and Attari scheme [36], A and B execute $4T_h$, $3T_c$ operations. Hence the total cost is $8T_h + 6T_c = 19,44,01,086$ ns. S executes $4T_h$ function, four-time T_c operations, hence the total cost is $4T_h + 4T_c =$

Table 7
Computation costs of different schemes in nanoseconds.

Schemes	Total execution cost
Zhao et al. [39]	12,96,02,674.396
Yang et al. [38]	51,05,836.576
Islam [37]	12,96,01,560.432
Farash and Attari [36]	21,67,33,142
Islam and Basu [2]	5395.35
Our's	4913.367

2,23,32,056.36 ns. Therefore the total execution cost of Farash and Attari scheme [36] is $19,44,01,086 + 2,23,32,056.36 = 21,67,33,142$ ns.

In Islam's scheme [37], both A and B are executed four hash operations, two times T_c operations, hence the total cost is $8T_h + 4T_c = 12,96,01,447.712$ ns. The S executes $8T_h$ functions. Therefore, the total communication cost of Islam scheme [37] is $12,96,01,447.712 + 112.72 = 12,96,01,560.432$ ns.

In Yang et al.'s scheme [38], both the users execute $5T_{exp}$ operations in G_t and two hash functions. Hence, final execution cost is $10T_{exp} + 4T_h = 34,37,723.856$ ns, and S executes $4T_{exp}$ in G_t and $8T_h$, hence the total cost is $4T_{exp} + 8T_h = 16,68,112.72$ ns. Therefore, the total execution cost of H. Yang et al. scheme [38] is $34,37,723.856 + 16,68,112.72 = 51,05,836.576$ ns.

In Zhao et al.'s scheme [39], both the users execute seven-time T_h functions and two T_c operations hence the total cost is $14T_h + 4T_c = 12,96,02,533.496$ ns, and S executes $10T_h$ functions. Hence, total communication cost of F. Zhao et al. protocol [39] is $12,96,02,674.396$ ns.

5.2. Communication cost

This section, we calculates the communication cost of our proposed scheme and existing schemes. We consider the size of ID_A and T_A to be 32 bits, x_i is 4096 bits, and the output of the g function to be 1 bits. In our proposed scheme the user A starts the communication and sends the login request message $\langle ID_A, ID_B, x_A, T_A, \Sigma_1 \rangle$ to the user B , where the size of ID_A is 32 bits, T_A is 32 bits, x_A is 4096 bits, Σ_1 512 bits, B sends the next message $\langle ID_A, ID_B, x_A, x_B, w_B, T_A, \Sigma_1, \Sigma_2 \rangle$ to the S , where the size of ID_A, ID_B, T_A are 32 bits each, x_A, x_B are 4096 bits, w_B 1 bit, and Σ_1, Σ_2 both are 512 bits. Next, the S sends the $\langle \Sigma_3, \Sigma_4 \rangle$ to B , where the size of Σ_3, Σ_4 are 512 bits. At the end, the B sends the message $\langle x_B, w_B, \Sigma_5 \rangle$ to the A , where the size of x_B is 4096 bits, w_B is 1 bit, Σ_5 is 512 bits. Therefore, the total communication cost is $4640 + 9313 + 1024 + 4641 = 19618$ bits per session.

In Islam and Basu's scheme [2], A and B sends the request message $\langle ID_A, T_A, x_A, \Sigma_A \rangle$, and $\langle ID_B, T_B, x_B, \Sigma_B \rangle$ to the S , where the size of

Table 8
Security attributes comparison of some 3PAKA schemes.

Security attributes\Schemes	[39]	[38]	[37]	[36]	[2]	Our scheme
Mutual Authentication	✓	✓	✓	✓	✓	✓
Known session Key security	×	✓	✓	×	✓	✓
Proposed secure password change phase	×	×	✓	✓	✓	✓
Off-line Password Guessing Attack	×	✓	×	✓	×	✓
Replay attack	×	✓	✓	×	✓	✓
Protect against Impersonation Attack	✓	✓	✓	✓	✓	✓
Avoid to use the public key of the server	✓	✓	✓	✓	✓	✓
Privileged Insider Attack	✓	✓	✓	×	✓	✓
Provably secure in Random Oracle Model	×	×	✓	✓	✓	✓
Secure against quantum attack	×	×	×	×	✓	✓

✓: Scheme prevents attack or satisfies the attribute; ×: Scheme fails to prevent attack or does not satisfy the attribute.

ID_A, T_A, ID_B, T_B are 32 bits each, x_A, x_B are 4096 bits each, Σ_A, Σ_B are 512 bits each. Next, S sends the response message $\langle ID_S, T_{S_A}, x_B, \Sigma_{x_A} \rangle$, and $\langle ID_S, T_{S_B}, x_A, \Sigma_{x_B} \rangle$ to A and B respectively, where the size of $ID_S, T_{S_A}, ID_S, T_{S_B}$ are 32 bits each, x_B, x_A are 4096 bits each, $\Sigma_{x_A}, \Sigma_{x_B}$ are 512 bits each. In the end, A and B exchange the message $\langle ID_A, w_A, \alpha_A \rangle, \langle ID_B, w_B, \alpha_B \rangle$ to each other, where the size of ID_A, ID_B are 32 bits each, w_A, w_B are 4096 bits each, α_A, α_B are 512 bits each. Therefore the total communication cost of Islam and Basu scheme [2] is $4672 + 4672 + 4672 + 545 + 545 = 19778$ bits per session.

In Farash and Attari's scheme [36], A sends the message $\langle ID_A, ID_B, R_A \rangle$ to S , where the size of ID_A, ID_B both are 32 bits, R_A is 256 bits, S sends the message $\langle ID_A, R_A, R_{S2} \rangle$ to B , where the size of ID_A is 32 bits, R_A, R_{S2} are 256 bits. Next, B sends the message $\langle R_B, Z_{BS}, Z_{BA} \rangle$ to S , where the size is 256 bits, and Z_{BS}, Z_{BA} are 512 bits. Now, S sends a message $\langle R_{S1}, R_B, Z_{BA}, Z_{SA} \rangle$ to A , where the size of R_{S1}, R_B are 256 bits each, and Z_{BA}, Z_{SA} are 512 bits each. A sends the message $\langle Z_{AS}, Z_{AB} \rangle$ to S , where the size of Z_{AS}, Z_{AB} are 512 bits each. Next, S sends a message $\langle Z_{AB}, Z_{SB} \rangle$ to B , where the size of Z_{AB}, Z_{SB} are 512 bits each. Therefore the total communication cost of Farash and Attari scheme [36] is $96 + 1536 + 4096 = 5728$ bits.

In Islam's scheme [37], A sends a request $\langle ID_A, request \rangle$ to B , and sends a message $\langle ID_A, Z_A \rangle$ to S , where the size of ID_A is 32 bits, Z_A is 128 bits which is an AES-256 encryption output, and B sends a request $\langle ID_B, request \rangle$ to A , and sends a message $\langle ID_B, Z_B \rangle$ to S , where the size of ID_B is 32 bits, and Z_B is 128 bits. Next, S sends a message $\langle ID_S, Z_{S_A} \rangle$ to A , and $\langle ID_S, Z_{S_B} \rangle$ to B , where the size of Z_{S_A}, Z_{S_B} both are 128 bits, and A sends a message $\langle ID_B, L_{AB} \rangle$ to B , and B sends a message $\langle ID_A, L_{AB} \rangle$ to A , where the size of L_{AB} , is 128 bits. Therefore, the total communication cost of Islam scheme [37] is $320 + 768 = 1088$ bits.

In Yang et al.'s scheme [38], A sends a message $\langle ID_A, V_A, Auth_A \rangle$ to B , where the size of ID_A is 32 bits, V_A is 1024 bits, $Auth_A$ is 512 bits, B sends a message $\langle ID_A, V_A, Auth_A, ID_B, V_B, Auth_B \rangle$ to S , where the size of ID_A, ID_B are 32 bits each, V_A, V_B are 1024 bits, $Auth_A, Auth_B$ is 512 bits, S sends a message $\langle ID_S, ID_B, w_A, R_B, Auth_A \rangle$ to A , and sends a message $\langle ID_S, ID_A, w_B, R_A, Auth_B \rangle$ to B , where the size of ID_S, ID_B, ID_A are 32 bits each, w_A, w_B are 32 bits each. Therefore, the total communication cost is $192 + 4096 + 3702 = 7990$ bits.

In Zhao et al.'s scheme [39], A sends a message $\langle R_A, W_A \rangle$ to S , and $\langle T_A, R_A \rangle$ to B , where the size of T_A is 32 bits, R_A is 256 bits, W_A is 256 bits, next, B sends a message $\langle R_A, W_A \rangle$ to A , and $\langle T_B, R_B, C_{BA} \rangle$ to S , where the size of C_{BA} is 512 bits, T_B is 32 bits, R_B is 256 bits. Next, S sends a message $\langle C_{S_A} \rangle$ to A and sends a message $\langle C_{S_B} \rangle$ to B , and A sends a message $\langle C_{AB} \rangle$ to B , where the size of C_{S_A}, C_{S_B}, C_{AB} are 512 bits each. Therefore, the total communication cost of Zhao et al. scheme [39] is $64 + 1536 + 2048 = 3648$ bits.

From Table 7, Table 9, and Table 8, it is clear that our protocol is better than Islam et al. protocol in terms of computation and communication cost and cryptographic security attributes.

Table 9
Communication costs of different schemes in bits per session.

Schemes	Communication cost
Farash and Attari [36]	5728
Islam [37]	1088
Yang et al. [38]	7990
Zhao et al. [39]	3648
Islam and Basu [2]	19778
Our's	19618

6. Conclusion

We propose an efficient three-party lattice-based protocol for a session key negotiation. This protocol resists the password-guessing attack. In ROM, our protocol is demonstrably safe according to the RLWE hardness assumption. Utilizing the Lattice Crypto Library, we estimate the execution and communication costs of our three-party authentication scheme for mobile devices. Moreover, a comparative study indicates that the proposed scheme's efficiency is improved. In the future, one may enhance the computation and communication overhead without compromising security. One may also consider the environment where mutual authentication can achieve without a trusted third party.

CRedit authorship contribution statement

Purva Rewal: Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing – review & editing. **Mrityunjay Singh:** Conception and design of study, Writing – original draft. **Dheerendra Mishra:** Conception and design of study, Writing – original draft. **Komal Pursharthi:** Acquisition of data, Analysis and/or interpretation of data, Writing – review & editing. **Ankita Mishra:** Acquisition of data, Writing – original draft.

Declaration of competing interest

The authors declare that the article entitled “Quantum-safe Three-Party Lattice based Authenticated Key Agreement Protocol for Mobile Devices” does not have any conflicts of interest. It is being declared that all the authors do not have any personal relationships with other people or organizations that could inappropriately influence (bias) their work.

Data availability

No data was used for the research described in the article

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